

Physical Inventory Counting

Front Royal, VA cold storage · Zebra MC9000-series handhelds

This is the working manual. It assumes you have never opened the app before and takes you from an empty deployment through to a finished count sent to the ERP. Read Part 1 and Part 2 once, when the app is first set up. Parts 3 to 5 are what you do for every count. Part 6 is the cycle-count programme, Part 7 is what to do when something goes wrong.

About the screenshots. Every picture below is the real application, photographed against a count part-way through a morning: the actual Front Royal bin list, three teams, four scanners, two aisles finished and three being counted. The names, pallet IDs and lot codes are made up. If a screen changes, `npm run docs` rebuilds every picture in this manual from scratch and re-prints [SOP.pdf](#) — the same manual typeset for paper.

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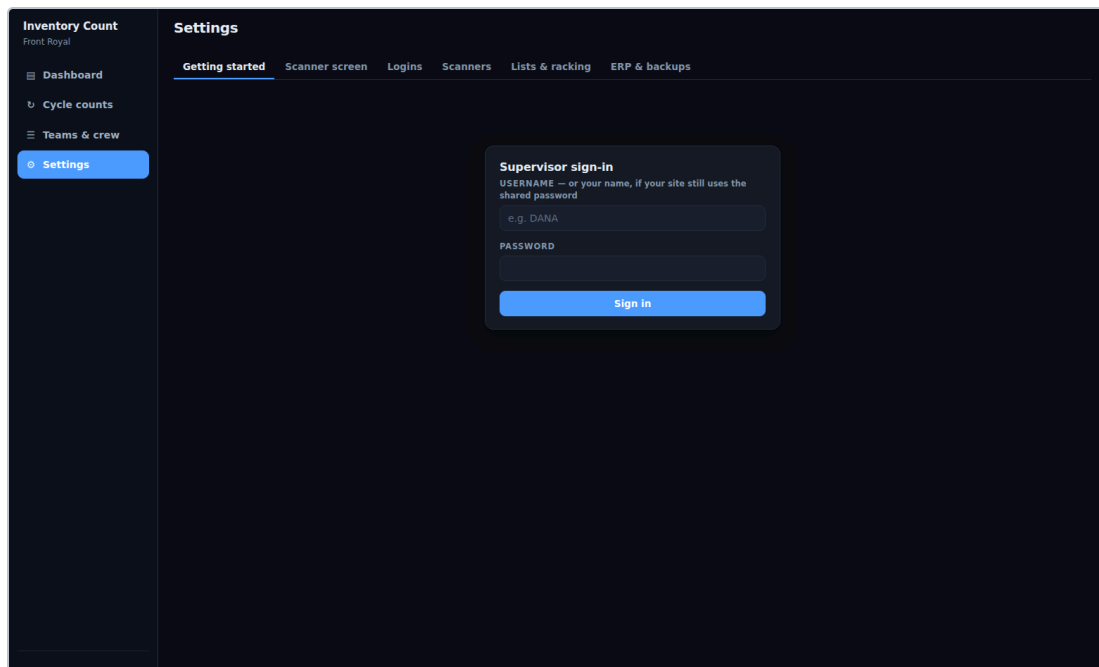
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Part 0 — What the system is

One server, four screens. Everything below happens in a browser; nothing is installed on a PC.

SCREEN	ADDRESS	WHO USES IT	SIGN-IN
Scanner	/	Counters, on the handhelds	Each scanner has its own link
Dashboard	/admin	Supervisor running the count	Supervisor login
Settings	/settings	Whoever sets the count up	Supervisor login
Office board	/board	Anyone — put it on the office TV	None. Read-only
Cycle counts	/cycle	Whoever runs the daily programme	Supervisor login



Every supervisor page opens on the same sign-in. Signing in on one carries to all four.

What the system does. It holds a list of every bin in the warehouse and a list of what the ERP thinks is in them. Counters scan pallets into bins on the handhelds. The app compares the two, tells you where they disagree, sends people back to look again where it matters, and produces an adjustment file for the ERP.

What a "count session" is. One count — a wall-to-wall in October, a spot check of the freezer, the year's cycle-count programme. Everything (bin list, inventory report, team assignments, every scanned line) belongs to a session. You can have several open at once; scanners choose which one they are working on at sign-on.

Two words you need.

- **Aisle** — a rack run, e.g. `F01`. Teams are assigned whole aisles.
- **Block** — aisles that back onto each other share a **block**, and only one team may be active in a block at a time, so two crews never work opposite faces of the same racking. Pairing aisles into blocks is a one-time job (Part 2, step 7).

Bin codes. A code like `F01A001` reads as zone `F`, aisle `01`, level `A`, position `001`. Four positions per bay; odd positions are the front face, even are the back. The app derives the aisle from the code when your file has no aisle column.

Part 1 — First-time setup: getting in

Do this once, with whoever administers the server. Allow 20 minutes.

Step 1 — Deploy the server

On Railway (how this site runs). Deploy the repository; `railway.json` builds the Dockerfile, and the app listens on the port Railway gives it. Attach a **persistent volume** and point `DB_PATH` at it — see step 2 — or the database is wiped on every redeploy.

On a PC inside the warehouse (no internet needed).

```
ADMIN_PASSWORD='pick-something' docker compose up -d
```

Then give that PC a static IP or a DHCP reservation, because the handhelds are pointed at its address and that address must never change. Scanners reach it at `http://<that-ip>:3000/`.

Step 2 — Set the server settings

On Railway these are **Variables**; on a PC they go in `docker-compose.yml`. Set these before anybody signs in:

SETTING	SET IT TO	WHY
<code>ADMIN_PASSWORD</code>	A password only supervisors know	The shared way in, used once to create real logins
<code>DB_PATH</code>	A path on the persistent volume, e.g. <code>/data/inventory.db</code>	Otherwise a redeploy loses the count
<code>SITE_TIMEZONE</code>	<code>America/New_York</code>	Dates on reports and cycle-count due dates
<code>SUPERADMIN_USER</code>	e.g. <code>sitelead</code>	Creates one permanent admin login at start-up
<code>SUPERADMIN_NAME</code>	That person's name, as it should appear in the log	Shown against everything they do
<code>SUPERADMIN_PASSWORD</code>	A strong password, at least 8 characters	If unset it falls back to <code>ADMIN_PASSWORD</code>

The superadmin account is created when the server starts and cannot be deleted from inside the app, so you can never lock yourself out. The app refuses to create it if the password would be the default `changeme` or is shorter than 8 characters — it will say so in the start-up log and carry on without it.

Never put a password in the repository. These are server settings for a reason: anything committed to git is readable by anyone with access to the code, for ever.

Step 3 — Sign in for the first time

Open `/settings`. Sign in with the shared `ADMIN_PASSWORD` (leave the username box empty), or with the superadmin username and password if you set one.

Step 4 — Create a login for each supervisor

Settings → Logins → Supervisor logins.

1. Type the person's **name**, a **username** and choose their **role**:
 - **Admin** — can do everything, including managing logins.
 - **Supervisor** — runs counts; cannot add or remove logins.
2. Press **Add**. The app shows a **starter password** once — something like `winter-4k2p`. Write it down and hand it to them; it is not shown again.
3. The first time they sign in they are made to choose their own password before they can do anything. Nobody but them knows it after that.

If somebody forgets their password, an admin presses **Reset password** on their row and gives them the new starter password. The same forced change happens again.

Supervisor logins

shared password only

everyone who changes something is recorded by name in the log

The shared password is the only way in.

Add an admin login below, sign in as it to check it works, then set SHARED_PASSWORD_LOGIN=off.

Give each supervisor their own login. **Admins** can manage accounts and everything else; **supervisors** can do everything except manage accounts. Deactivating a login keeps its history in the log but stops it being used.

USERNAME	FULL NAME	PASSWORD	ROLE	
DANA	Dana Whitfield	leave blank to generate	Supervisor	Add login

Leave the password blank and one is generated for you — read it out once, and they choose their own the first time they sign in. Type one instead and they still have to replace it, so nobody but its owner ever knows a working password.

USERNAME	NAME	ROLE	STATUS	LAST SIGNED IN	ADDED
No logins yet — everyone is signing in with the shared password.					

Your password

You are signed in with the shared password, which has no account to change. Ask an admin for a login of your own.

Settings → Logins. Each person has their own login and their own password; the starter password is shown once.

Step 5 — Turn the shared password off

Once at least one admin login exists, set `SHARED_PASSWORD_LOGIN=off` in the server settings. From then on everybody signs in as themselves and the log names who did what.

The app will not let this lock you out: if there are no admin accounts, the shared password keeps working regardless of the setting.

Part 2 — First-time setup: the warehouse

Also once, though you will come back to steps 6 and 7 whenever the racking changes.

Step 6 — Upload the bin list

Settings → Lists & racking → Bin list.

This is every location in the warehouse. It is what a scanned bin is checked against, and it is what defines the aisles teams get assigned to. **Upload this before anything else.**

- For this site, press **Load the Front Royal bin list** — the ERP bin export ships with the app: racks F01-F24 and A01-A04 plus WIP, the NIL bin and other areas. Staging lanes and dock doors are counted manually and are deliberately left out.
- Otherwise choose a CSV or Excel file. Column names are matched loosely — see [Appendix A](#). The only column you must have is the bin location.
- **Replace what is there** wipes the existing list first. Leave it unticked to add to it.

The card reports what it read: how many bins, how many aisles, and anything it skipped.

Bin list every location in the warehouse

What a scanned bin is checked against, and what defines the aisles teams get assigned to. Upload this one first.

CSV OR EXCEL FILE

No file chosen ☐ Replace what is there

Columns — matched by name, any order, extra columns ignored. [Download a sample bins-template.csv](#)

Bin Location (required)	Aisle	Zone	Description
-------------------------	-------	------	-------------

Bin Location: the code on the bin label; also accepts Location, Bin, Slot · **Aisle:** which aisle the bin is in - taken from the first part of the code if missing (A03-12-1 → A03) · **Zone:** area of the warehouse, for the progress view

The ERP bin export that ships with the app: racks F01-F24 and A01-A04 plus WIP, the NIL bin and other areas. Staging lanes and dock doors are counted manually and are left out.

Settings → Lists & racking. Each list has its own card, and each reports exactly what it read.

Step 7 — Pair the aisles into racking blocks

Settings → Lists & racking → Aisles & racking blocks.

Aisles that back onto each other must share a block so two teams are never working the same racking from both sides.

1. Set **Aisles per block** to 2 (most double-deep racking).
2. **Skip first** is 1 if the first aisle has a wall behind it rather than another aisle.

3. Press **Auto-pair aisles**, then read the table and correct anything that is wrong by typing a block name straight onto a row.

If you have uploaded a rack drawing, **Pair from drawing** uses it instead — more reliable than counting aisles off by hand.

Aisles & racking blocks

Aisles in the same **block** share racking, so only one team can be active in a block at a time. Pair them automatically, or type a block name on any row. Teams are queued onto these aisles on the [Dashboard](#).

AISLES PER BLOCK

2

SKIP FIRST

0

Auto-pair aisles

Pair from drawing

AISLE	BLOCK	BINS	COUNTED	PROGRESS	STATUS
Dry Dock - Aisle A01	A01+A02	132	0		—
Dry Dock - Aisle A02	A01+A02	258	0		—
Dry Dock - Aisle A03	A03+A04	258	0		—
Dry Dock - Aisle A04	A03+A04	36	0		—
Freezer - Aisle F01	F01+F02	654	652		done
Freezer - Aisle F02	F01+F02	618	240		active team 1 (ABCDEF)
Freezer - Aisle F03	F03+F04	618	617		done

Aisles paired into racking blocks. F01+F02 share racking, so only one team works that block at a time.

Step 8 — Register the scanners

Settings → Scanners → Scanner setup.

Each handheld gets its own link. Opening that link once is what signs the scanner in; after that the server accepts counts from it and stamps its name on every line.

1. Type a **scanner name** — use what is written on the device, e.g. `SCANNER-05`. Add notes (asset tag, "freezer unit") if it helps.
2. Press **Add scanner**. Repeat for every handheld.
3. Press **Print setup cards**. You get one card per scanner with its own QR code. Cut them up and tape one inside each cradle.

Scanner setup scanners must sign in each scanner gets its own link — open it once on the device and add it to the home screen

SCANNER NAME: NOTES (OPTIONAL):

[Add scanner](#) [Print setup cards](#)

SCANNER	LINK			SIGNED IN	LAST SEEN	TEAM	NOTES		
SCANNER-01	http://127.0.0.1:3399/?d=ajpbgxzr	Copy	QR	9/16/2026	9/16/2026, 5:42:50 PM	1	freezer unit A	Reset link	Remove
SCANNER-02	http://127.0.0.1:3399/?d=zav343ed	Copy	QR	9/16/2026	9/16/2026, 5:42:50 PM	2	freezer unit B	Reset link	Remove
SCANNER-03	http://127.0.0.1:3399/?d=8s3jkjuf	Copy	QR	9/16/2026	9/16/2026, 5:42:50 PM	3	freezer unit C	Reset link	Remove
SCANNER-04	http://127.0.0.1:3399/?d=h2dd7gtp	Copy	QR	9/16/2026	9/16/2026, 5:42:50 PM	—	spare, in the office	Reset link	Remove

On the device: open the link in Chrome (or scan the QR into the address bar), then menu → **Add to Home screen**. Opening the link signs that scanner in — the server accepts counts from signed-in scanners only. Treat a link like a key: **Reset link** issues a new one and kills the old, and removing a scanner stops it entirely. A link used more than once is flagged, which is normal after a scanner is wiped and worth a look otherwise.

Settings → Scanners. One row per handheld: its link, when it was last seen, Reset link and Remove.

On each handheld, once:

1. Open Chrome, scan the QR code from its card into the address bar (or type the link).
2. Chrome menu → **Add to Home screen** (or tap **Install on this scanner** on the app's own sign-on screen). **Do this — do not leave it running as a browser tab.** Installed, it launches with no browser around it and starts even with no signal; left as a tab, Chrome slides its address bar over the counting screen every time the page moves.
3. The app takes the whole screen once a counter signs on. For a device that should do nothing else at all, lock it down with Zebra's **Enterprise Home Screen**, pointing it at this app — that is a device setting, not an app one.
4. Set up **DataWedge** so scans arrive as keystrokes: sees the end of a scan)*
 - the profile associated with Chrome → **Keystroke output: enabled**
 - **Basic data formatting → Send ENTER key: enabled** *(without this the app never
 - **Barcode input** → enable the symbologies your labels use.

Treat a scanner link like a key. **Reset link** issues a new one and kills the old — use it if a device is lost or a link gets out. Removing a scanner stops it entirely. A link used on more than one device is flagged in the table; that is normal after a scanner is wiped and worth asking about otherwise.



Print setup cards: one per scanner, each with its own QR code and the three steps to do on the device.

Step 9 — Set up the counting screen

Settings → Scanner screen.

This is what a counter actually sees. The preview on the right is drawn at the real pixel size of the device (MC9090 is 240 × 320; MC9200 is 480 × 640) using the scanner's own stylesheet, so it cannot drift from reality.

- **The questions, in order.** Drag to reorder. The default asks for the pallet, then the quantity, then the bin. Counting **location-first** — bin, then what is in it — suits a team working a bay at a time. Lot code and expiry sit in the order too, and only appear on counts that ask for them (Part 3). A question can never be dragged off the list entirely.
- **Show contents after a pallet scan** — shows the SKU and description the ERP has for that pallet, so the counter can see they are at the right one.
- **Show the next bin in the aisle** — once a team starts a section, the gun tells them which bin comes next (001 → 002 → 003), so nothing is skipped. A bin holding several pallets keeps the guide until every tag in it is accounted for — see step 15.
- **Take the whole screen** — off unless you turn it on. It makes the app fill the display at sign-on, but the browser announces that with a bar carrying the site's address, right over the counting screen. **Installing the app** on the handheld (step 8) is the quiet way to lose the browser, and the one to prefer.
- **Keep the screen awake** — holds the display on while a team is counting, instead of it sleeping between bays.
- **Buzz on a good or bad scan** — useful with ear defenders.

- **Text size: Large** — for gloves and a freezer.
- **Re-key a quantity of at least** — anything this big has to be typed twice. 0 never asks twice.

Press **Save**. Scanners pick the change up within about half a minute, between pallets — nobody has to sign out.

How the counting screen is put together the defaults

The order the questions are asked, and how much the gun shows. Changes reach a scanner within about half a minute. The preview is drawn at the real screen size, so what you see is what a counter sees.

The questions, in order
 Drag to reorder. Counting location-first — bin, then what is in it — suits a team working a bay at a time.

- Pallet ID**
scan the label on the pallet or container
- Quantity**
how many are on it
- Lot code**
only on counts that track lots
- Expiry**
only on counts that track best-before dates
- Bin**
scan the location label

What the gun shows

- ☒ Show contents after a pallet scan
- ☒ Show the next bin in the aisle
- ☒ Buzz on a good or bad scan
- ☒ Take the whole screen
- ☒ Keep the screen awake

TEXT SIZE
 Normal

RE-KEY A QUANTITY OF AT LEAST
 1000 · 0 never asks twice

Save **Restore the defaults**

PREVIEW ON
 MC9090 — 240 × 320

SHOWING
 Pallet ID (step 1)

Team 1 · F01 0 queued

NEXT BIN
F01A001
 0 of 94

STEP 1 OF 6
Scan PALLET ID

Keyboard **Back**

MC9090 · 3.8" QVGA · 240 × 320

Settings → Scanner screen. Drag the questions into the order your site counts in; the preview is the real MC9090 screen size.

Step 10 — Set the one-tap reasons

Settings → Scanners → What the scanners offer.

Counters wearing gloves in a freezer will not type. These are the buttons they tap instead, and the reasons a site needs are its own — "Blocked by a trailer" means something here and nothing anywhere else.

- **Comments** — offered on the last step, after the bin is scanned.
- **Override reasons** — offered when a pallet ID is not on the list and the counter is allowed to accept it anyway. **Other** is always offered on the gun as well, whatever you configure.
- **Comments step moves on after N seconds** — the gun counts down and then moves to the next bin by itself. Typing or tapping a chip stops the clock. Set it to 0 to make it wait for the counter.

Press **Save**; again, scanners pick it up within about half a minute.

What the scanners offer

the defaults

the one-tap reasons a counter can pick, without typing

The reasons a site needs are the site's own — "Blocked by a trailer" means something here and nothing anywhere else. A scanner picks up a change within about half a minute, without anybody signing out of it.

Comments

Offered on the last step, after the bin is scanned.

Damaged

Partial pallet

Mixed pallet

Label unreadable

Needs recount

Blocked / could not reach

e.g. Frozen to the rack

Add

Override reasons

Offered when a pallet ID is not on the list and the counter is allowed to accept it anyway.

Label unreadable

New receipt, not on the report

Relabelled

Hand-written ID

Supervisor said to count it

e.g. Hand-written ID

Add

COMMENTS STEP MOVES ON AFTER

5

seconds · 0 waits for the counter

Save

Restore the defaults

Settings → Scanners → What the scanners offer. These are the buttons a counter taps instead of typing.

Physical Inventory Counting · SOP

13

Part 3 — Setting up a count

Do this the day before, not on the morning of the count.

Step 11 — Create the count

Dashboard → Progress → Count session → Start a new count.

1. **Name** it something you will recognise later: `Q3 2026 wall-to-wall`.
2. **Type**:
 - **Full count** — teams work whole aisles. This is a wall-to-wall.
 - **Cycle count** — a batch of bins per day or week; see Part 6.
3. **Bin list** — attach it here, or upload it later in Settings.
4. **Inventory report** — what the ERP thinks is on hand. **Attach it**. With it you get variances, second counts and an ERP adjustment file. Without it the count still records what is there, but nothing is compared against anything.
5. Press **Create count**.

The new count inherits the rack drawing the last one used, so the map works immediately.

The screenshot shows the 'Inventory Count' dashboard with a sidebar on the left containing 'Dashboard', 'Cycle counts', 'Teams & crew', and 'Settings'. The main area is titled 'Dashboard' and shows a 'Progress' section with a 'FULL Q3 2026 wall-to-wall 13%' status. Below this is a table with columns: TEAM, SCANNER(S), EMPLOYEES, ACTIVE AISLE, LINES, BINS, and LAST SCAN. The table has three rows of scanner data. Below the table is a 'Count session' section with various settings like 'Validate', 'Guided by aisle plan', and 'Ask for the expiry date'. At the bottom is a 'Start a new count' form with fields for NAME, TYPE, BIN LIST, and INVENTORY REPORT, and a 'Create count' button.

TEAM	SCANNER(S)	EMPLOYEES	ACTIVE AISLE	LINES	BINS	LAST SCAN
1	SCANNER-01	E1043, E1088	Freezer - Aisle F02	1151	892	5:39:40 PM
2	SCANNER-02	E1102, E1157	Freezer - Aisle F04	1027	797	5:33:40 PM
3	SCANNER-03	E1163, E1190	Freezer - Aisle F05	178	120	5:41:40 PM

The count is picked in the header, on every supervisor page. Each row carries its own progress, so two counts are never confused.

Step 12 — Check the guided setup

Settings → Getting started.

A checklist worked out from what is actually in the database, not from a box somebody ticked. It tells you what is still missing and takes you straight to the screen that fixes it. For a full count it wants: the bin list, the inventory report, registered scanners, the scanner screen, racking blocks, a crew roster and a team plan.

Getting a count started October spot check · full count
Worked out from what is actually in the database, not from a box somebody ticked. Pick the count at the top of the page; the steps change for a cycle count.

2 thing(s) still needed before anyone can scan.
The rest is recommended rather than required.

- ! Upload the bin list** needed to start [Upload it](#)
Every location in the warehouse. It is what a scanned bin is checked against, and it defines the aisles teams get assigned to.
nothing uploaded yet
- Upload the inventory report** recommended [Upload it](#)
What the system thinks is on hand. Without it a count still records what is there, but nothing is compared, so there are no variances and no second counts.
nothing uploaded yet
- ✓ Register the scanners** [Change it](#)
Each handheld gets its own link. The server only takes counts from a scanner that has signed in with one.
4 registered · 4 signed in
- ✓ Pick the rack drawing** [Change it](#)
Draws progress over the real floor plan instead of a schematic.
using front-royal
- Pair the aisles that share racking** recommended [Pair them](#)
Two teams either side of the same rack get in each other's way. Pairing them keeps only one team in a block at a time.
every aisle is its own block
- ✓ Add the crew and the teams** [Change it](#)
Lets the app check a team has the equipment for the levels it is given, and check who signed on.
8 on the crew list · 3 team(s)
- ! Queue the aisles onto teams** needed to start [Queue them](#)
A guided count sends each team to its next aisle. Without a plan the scanners have nothing to hand out.
nothing queued yet

A new count: two things stop anybody scanning, the rest are recommended. Every step links to the screen that fixes it.

Getting a count started Q3 2026 wall-to-wall · full count
Worked out from what is actually in the database, not from a box somebody ticked. Pick the count at the top of the page; the steps change for a cycle count.

This count is running — 1,809 bins have a count already.
Anything still open below can be filled in as you go.

- ✓ Upload the bin list** [Change it](#)
Every location in the warehouse. It is what a scanned bin is checked against, and it defines the aisles teams get assigned to.
13,673 bins in 28 aisles
- ✓ Upload the inventory report** [Change it](#)
What the system thinks is on hand. Without it a count still records what is there, but nothing is compared, so there are no variances and no second counts.
2,658 pallets · 1,809 bins carry a last-counted date
- ✓ Register the scanners** [Change it](#)
Each handheld gets its own link. The server only takes counts from a scanner that has signed in with one.
4 registered · 4 signed in
- ✓ Pick the rack drawing** [Change it](#)
Draws progress over the real floor plan instead of a schematic.
using front-royal
- ✓ Pair the aisles that share racking** [Change it](#)
Two teams either side of the same rack get in each other's way. Pairing them keeps only one team in a block at a time.
14 block(s) of more than one aisle
- ✓ Add the crew and the teams** [Change it](#)
Lets the app check a team has the equipment for the levels it is given, and check who signed on.
8 on the crew list · 3 team(s)
- ✓ Queue the aisles onto teams** [Change it](#)
A guided count sends each team to its next aisle. Without a plan the scanners have nothing to hand out.
8 aisle assignment(s)

The same checklist once that count is ready to run: every step green, and the count already running.

Step 13 — Set the count's options

Dashboard → Progress → Count session. These apply to the count picked in the header, and a scanner that is already counting picks up a change within about half a minute. The questions it asks change between pallets, never mid-line.

OPTION	WHAT IT DOES	SUGGESTED
Pallet ID check	<i>Validate</i> — allow override with reason: an unknown pallet can be accepted with a reason. <i>Validate</i> — no overrides: it cannot. <i>Accept any ID</i> : no checking, though duplicates are still blocked.	Validate with override
Guided by aisle plan	Teams are sent to their assigned aisle and warned when they scan a bin outside it. Off means anyone can count anything.	On for a wall-to-wall
Ask for comments	Adds the optional comments step at the end of each pallet.	On
Auto second counts	Raises a "go back and look again" task automatically when a line disagrees with the report.	On
Scanners start here	Every scanner lands on this count at sign-on. They can still pick another.	On, on count day
Ask for the lot code	Adds a LOT CODE question, checked against the report. Wrong lot is called out at the pallet.	On only if you track lots
Ask for the expiry date	Adds an EXPIRY question and flags anything already out of date.	On for frozen food
Recount over N units	A difference smaller than this raises no second count.	2-5 units
or over N %	...or at least this share of the expected quantity. Either threshold is enough.	5-10 %
Cap open	Stop raising automatic second counts once this many are open, so the list stays walkable.	50-100
Map drawing	Which rack drawing the map uses. <i>Schematic</i> builds one from the bin codes.	Your drawing

Why the thresholds matter. With both at 0 the app sends somebody back for a one-unit difference, and by mid-morning the second-count list is longer than the count. Set them once and the list stays short enough that people actually walk it.

Count session #1 · full count · open

Settings for the count picked in the header. A scanner already counting picks up a change within about half a minute; the questions it asks change between pallets, never mid-line. The **recount thresholds** decide how big a quantity difference has to be before somebody walks back to the bin — leave them at 0 to send someone back for every unit.

PALLET ID CHECK

Validate — allow override with reason ☒ Guided by aisle plan ☒ Ask for comments ☒ Auto second counts ☒ Scanners start here ☒ Ask for the lot code

RECOUNT OVER OR OVER CAP OPEN MAP DRAWING

☒ Ask for the expiry date 3 units 5 % 60 Front Royal - rack layout D-22P9170-L001 (rev A) (28 aisles) **Save** **Close session**

Delete...

Start a new count

NAME **TYPE**

e.g. Q3 2026 wall-to-wall Full count — teams work whole aisles

BIN LIST — optional, you can upload it later **INVENTORY REPORT** — what the count is validated against

Choose File No file chosen **Choose File** No file chosen **Create count**

The inventory report is what the system thinks is on hand. With it you get variances and second counts; without it a count still records what is there, but nothing is compared. Either file can wait — [Getting started](#) tracks what is still missing.

Dashboard → Progress → Count session: every option above, on one card, for the count picked in the header.

Step 14 — Give the teams their aisles

Dashboard → Team plan.

Either upload a **counting plan** (Settings → Lists & racking → Counting plan: one row per aisle, in the order each team counts it, with the levels their equipment reaches), or queue aisles by hand here.

- A team's next aisle starts automatically as soon as the block it belongs to is clear. That is what staggers the crews apart.
- A team is not sent where its equipment cannot reach: if the roster says a crew is on foot, they will not be queued onto level D. A supervisor can override this.
- Two teams **can** share one aisle as long as their levels do not overlap — a forklift crew high and a crew on foot low.

Team assignments

Queue aisles per team in counting order, with the **levels** that team's equipment covers (A-C on foot, D-F with the lift, A-F for all). Levels are required. Two teams can share an aisle on different levels. A team's next aisle waits while another team holds those levels in its racking block — that is what staggers them. Which aisles share racking is set under [Settings](#).

TEAM 3 AISLES, IN ORDER F01, F03, F05 LEVELS (REQUIRED) A-C [Queue](#)

Team 1 in Freezer - Aisle F02, levels A-F

Freezer - Aisle F01 - levels A-F [reopen](#) Freezer - Aisle F02 - levels A-F [done](#) [release](#) Freezer - Aisle F07 - levels A-F [start](#) [x](#)

Team 2 in Freezer - Aisle F04, levels A-F

Freezer - Aisle F03 - levels A-F [reopen](#) Freezer - Aisle F04 - levels A-F [done](#) [release](#) Freezer - Aisle F08 - levels A-F [start](#) [x](#)

Team 3 in Freezer - Aisle F05, levels A-F

Freezer - Aisle F05 - levels A-F [done](#) [release](#) Freezer - Aisle F06 - levels A-F [start](#) [x](#)

Dashboard → Team plan: each team's aisles in counting order — what they are on now, and what is queued behind it.

Inventory Count
Front Royal

Dashboard
Cycle counts
Teams & crew
Settings

Teams & crew 8 people · 3 teams

Crew & teams Equipment rules

Crew list clock in number, name, department, equipment
CSV OR EXCEL FILE

[Choose File](#) No file chosen ☐ Replace the whole list [Upload](#) [Sample CSV](#) [Export current list](#)

Columns matched by name: **Clock In Number** (also accepts Badge, Employee ID), then Name, Department, Equipment. Equipment can be a list — Scissor lift; High reach. Spellings like *forklift*, *reach truck*, *pallet jack* are understood.

CLOCK IN NUMBER	NAME	DEPARTMENT	EQUIPMENT
E1042	Full name	Freezer	☐ Dock truck ☐ Scissor lift ☐ High reach truck

[Add / update](#)

Teams drag people between the crew list and a team — or use the menu on a card

NEW TEAM 3 NOTES e.g. night shift, freezer [Add team](#)

Not on a team 2

Drag someone onto a team to assign them.

Nadia Haddad E1204 · Warehouse · reaches level A
On foot [Remove](#)

Owen Blackwell E1219 · Warehouse · reaches level A
Dock truck [Remove](#)

Team 1 [reaches levels A-F](#)

Between them: High reach truck, Scissor lift, Dock truck, On foot

Marcus Obi E1043 · Warehouse · reaches levels A, C, D, E, F
Team 1 [Remove](#)

High reach truck Scissor lift

Priya Raman E1088 · Warehouse · reaches level A
Team 1 [Remove](#)

Dock truck On foot

[Delete team](#)

Team 2 [reaches levels A-F](#)

Between them: Dock truck, On foot, High reach truck, Scissor lift

Ava Delgado E1157 · Warehouse · reaches level A
Team 2 [Remove](#)

Dock truck On foot

Tom Zielinski E1102 · Warehouse · reaches levels A, C, D, E, F
Team 2 [Remove](#)

High reach truck Scissor lift

[Delete team](#)

Team 3 [reaches levels A-F](#)

Between them: Dock truck, On foot, High reach truck, Scissor lift

[shared password](#) [Log out](#)

Teams & crew (/teams): the crew list and who is on which team. Equipment is what decides which levels a team can be sent to.

Part 4 — Counting day

Step 15 — The counter's procedure (on the handheld)

Signing on

1. Open the app from the home screen. It shows the scanner's own name.
2. Pick the **count session** (it lands on the default one).
3. Enter the **team number**.
4. Scan or type every **clock-in number** on the crew, pressing Enter after each. Tap a number to remove it.
5. Press **Sign on & load list**. The handheld downloads the bin and pallet lists for that count and can work from then on **with no signal**.

The screenshot shows the 'Inventory Count' app interface. At the top, it says 'SCANNER-01' and 'online'. Below that, it asks 'WHAT ARE YOU DOING?' with two buttons: 'Full count' (highlighted in blue) and 'Cycle count'. Then it asks 'WHICH ONE' with a dropdown menu showing '#1 — Q3 2026 wall-to-wall'. Below that, it asks 'TEAM NUMBER' with a text input field containing '1'. Then it asks 'CLOCK IN NUMBERS — SCAN OR TYPE, THEN ENTER' with a text input field containing 'Clock in number' and an 'Add' button. Below that, there are two buttons: 'E1043 x' and 'E1088 x'. Then it says 'Everyone on the crew — tap a number to remove it.' Below that, there are two buttons: 'Sign on & load list' (highlighted in blue) and 'Refresh sessions'. At the bottom, it says 'Change scanner ID' and 'This scanner is registered as SCANNER-01.'

Signing on: the count, the team number, and every clock-in number on the crew.

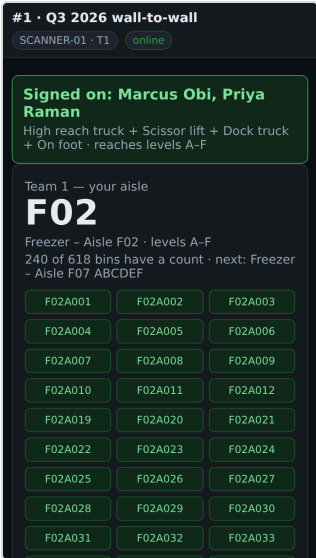
Counting a pallet

The gun asks one question per screen, in the order the site configured:

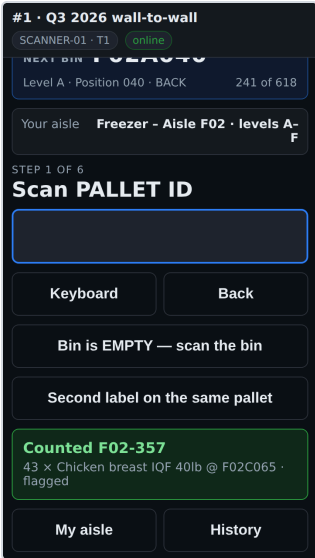
1. **Scan PALLET ID** — it shows what the ERP says is on that pallet.
2. **Enter QUANTITY** — big numbers may have to be typed twice.
3. **Scan LOT CODE / Enter EXPIRY** — only on counts that ask for them. A lot that disagrees with the report, or a date already past, is called out on the spot.
4. **Scan BIN LOCATION** — it says where that bin is and which face it is on.

5. **Comments** — optional. Tap a reason or type a note; leave it and the gun moves on by itself.

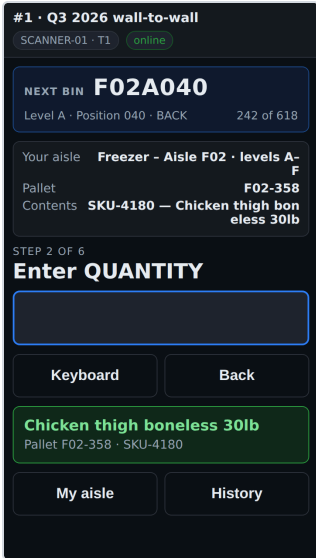
The line is saved on the handheld the moment the last question is answered, and pushed to the server whenever there is a signal. The header shows `online` / `OFFLINE` and how many lines are still queued.



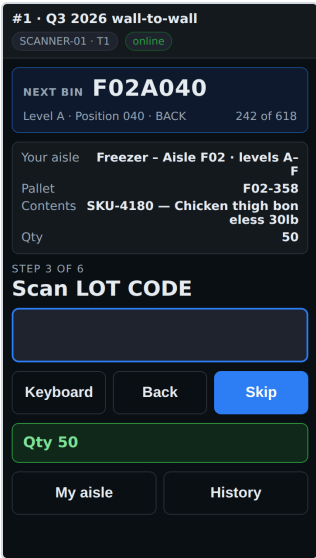
After sign-on — the aisle this team has been given, what is queued behind it, and any second counts waiting



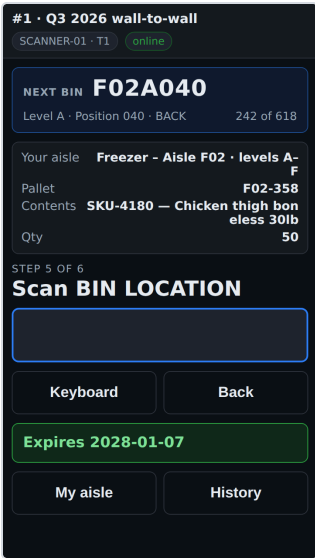
Question 1 — *Next bin* tells the team which bin comes next in the aisle, so nothing gets skipped



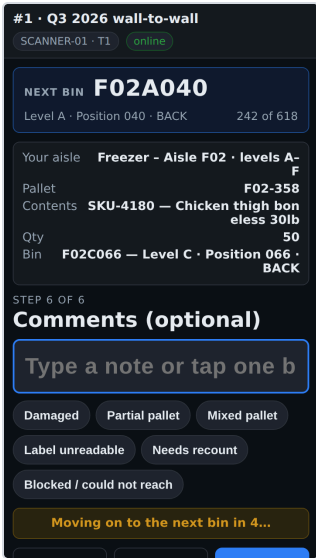
Question 2 — the gun shows what the ERP says is on that pallet, so the counter can see they are at the right one



Questions 3 and 4 — only on counts that ask for them. A lot that disagrees with the report is called out here, not a week later in a report



The last required question — the gun says where that bin is and which face it is on



Comments — tap a reason rather than type one. Left alone, the gun counts down and moves to the next bin by itself

A bin with more than one pallet in it

Four pallet tags in one position is ordinary, and the guide stays put until they are all counted. After the first tag the banner turns amber and reads **Still in this bin** with a tally — *2 of 4 tags* — taken from the inventory report, so the team knows what is left to find before they walk on.

- Scan the next tag in the same bin. The tally goes up; when the report's pallets are all accounted for, the banner turns back to **Next bin** on its own.
- If the report is wrong and there is nothing more there, tap **Nothing more here** →. The bin is closed and the guide moves on.
- A bin the report lists nothing for stays open after a count, for the same reason — there may be a second pallet in it that the ERP has never heard of. Tap **Nothing more here** → when it is clear.
- **Bin is EMPTY** closes a bin outright.

The tally counts what the whole team has done, not just this scanner, so two handhelds working the same aisle never double-walk a bay or leave one half counted.



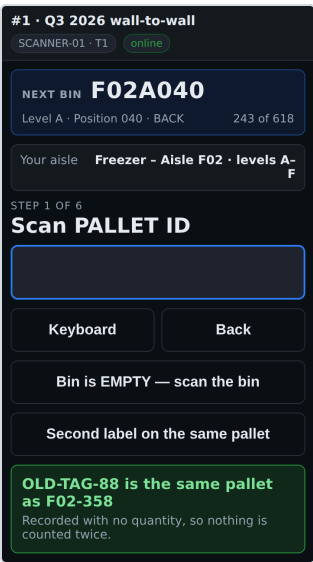
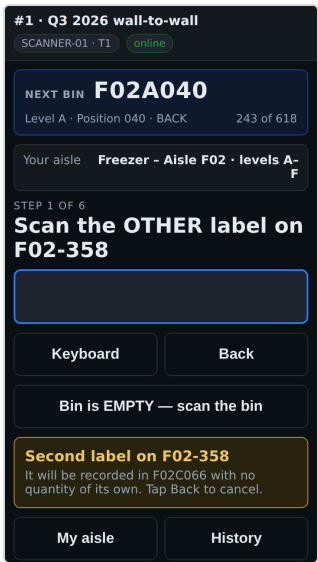
One tag into a position that holds three: the banner turns amber and stays on the bin until the other two are counted, or somebody says there is nothing more there.

A pallet with two labels on it

A re-tagged pallet that kept its old label, or one built from two, has two barcodes and is one pallet. Counting both as pallets doubles the stock; ignoring one leaves a live label for the next team to find.

1. Count the pallet once, as normal.
2. Tap **Second label on the same pallet**.
3. Scan the other label.

The tag is recorded against the pallet it is stuck to, with no quantity of its own. It shows on the pallet report as **SECOND LABEL**, the pallet it belongs to says *also tagged*, nobody is sent back for it, and it never reaches the ERP file as an adjustment. If the second tag is not on the inventory report, the gun offers the same answer in the "Pallet not on the list" screen.



The gun asks for the other label and names the pallet it will belong to

Recorded as the same pallet, with no quantity — so the stock is not counted twice

The other buttons

BUTTON	WHEN TO USE IT
Back	Wrong entry — steps back one question
Skip	Lot, expiry or comments the pallet does not have
Bin is EMPTY	The bin is genuinely empty. Scan the bin; it is recorded as counted and empty
My aisle	Back to the assignment screen
History	The last 50 lines from this scanner. Void removes a wrong line from the totals
Aisle complete — next aisle	Only when the aisle is finished. It frees the racking block and releases the team's next aisle

#1 · Q3 2026 wall-to-wall

SCANNER-01 · T1 online

Recent counts

Most recent 50 lines from this scanner. Void removes a line from the totals.

OLD-TAG-88 @ F02C066 5:44:10 PM · synced · second label on F02-358	0
F02-358 @ F02C066 5:44:06 PM · synced	50
F02-365 @ F02C070 5:44:01 PM · synced	59
F02-364 @ F02C070 5:43:58 PM · synced	52
F02-363 @ F02C070 5:43:54 PM · synced	45
F02-357 @ F02C065 5:43:51 PM · synced · flagged: expired 2026-08-27	43

History: the last 50 lines this scanner counted. **Void** takes a wrong one out of the totals.

When the gun asks for a reason

An unknown pallet, an unknown bin, a pallet already counted, a bin outside the team's aisle or level — the gun stops and asks why before it will take the line. Pick a reason from the list (or **Other**), add a note if it helps, and press **Accept and continue**. The line is saved and flagged for a supervisor. **Cancel — rescan** if it was simply the wrong barcode.

#1 · Q3 2026 wall-to-wall

SCANNER-01 · T1 online

Pallet not on the list

NO-LABEL-4471 is not in the uploaded pallet file.

Scanned **NO-LABEL-4471**

REASON (REQUIRED TO CONTINUE)

Choose a reason...

NOTE (OPTIONAL)

Free text

Accept and continue

It is a second label on F02-358

Cancel — rescan

#1 · Q3 2026 wall-to-wall

SCANNER-01 · T1 online

Pallet not on the list

NO-LABEL-4471 is not in the uploaded pallet file.

Scanned **NO-LABEL-4471**

REASON (REQUIRED TO CONTINUE)

Hand-written ID

NOTE (OPTIONAL)

Tag written by receiving

Accept and continue

It is a second label on F02-358

Cancel — rescan

A pallet that is not on the inventory report: the gun will not take the line until somebody says why

The reasons in that list are the ones set in Settings, so they are your site's words. **Other** is always offered

Step 16 — The supervisor's procedure (on the dashboard)

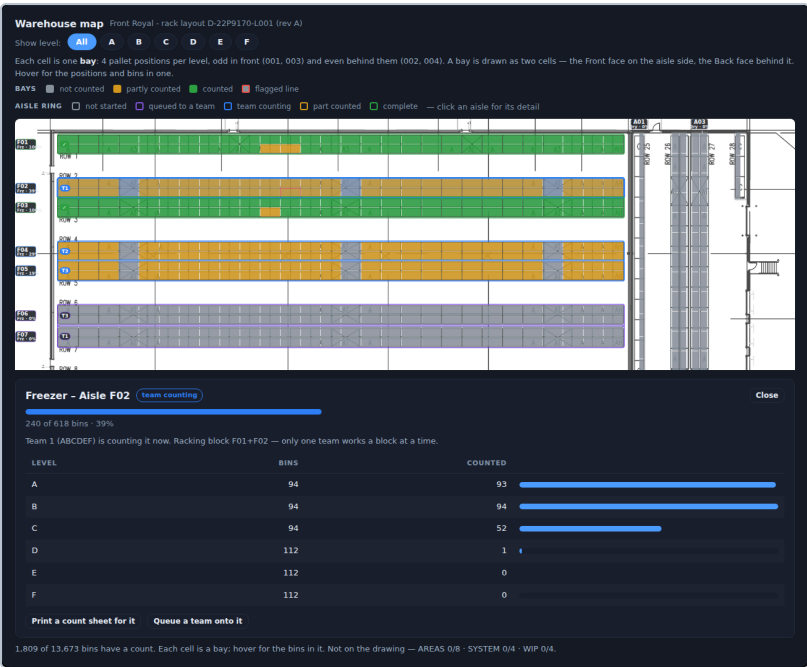
Dashboard, with the count picked in the header. It refreshes itself every 30 seconds.

- **Progress** — lines, bins counted, pallets, exceptions; and a row per team with what they are counting right now and when they last scanned. A team whose last scan was 40 minutes ago is a team with a problem.



Dashboard → Progress. The row per team is the one to watch: what they are on, how much they have done, and when they last scanned.

- **Map** — the warehouse from above. Aisles are outlined by state: not started, being counted now, done. Click an aisle for its bins, who counted it and what is flagged.



Dashboard → Map, with an aisle picked: 240 of 618 bins, level by level, and which team holds that racking block.

- **Team plan** — who is where, queue the next aisles, hand an aisle back.
- **Second counts** — see step 17.
- **Reports** — see step 18.

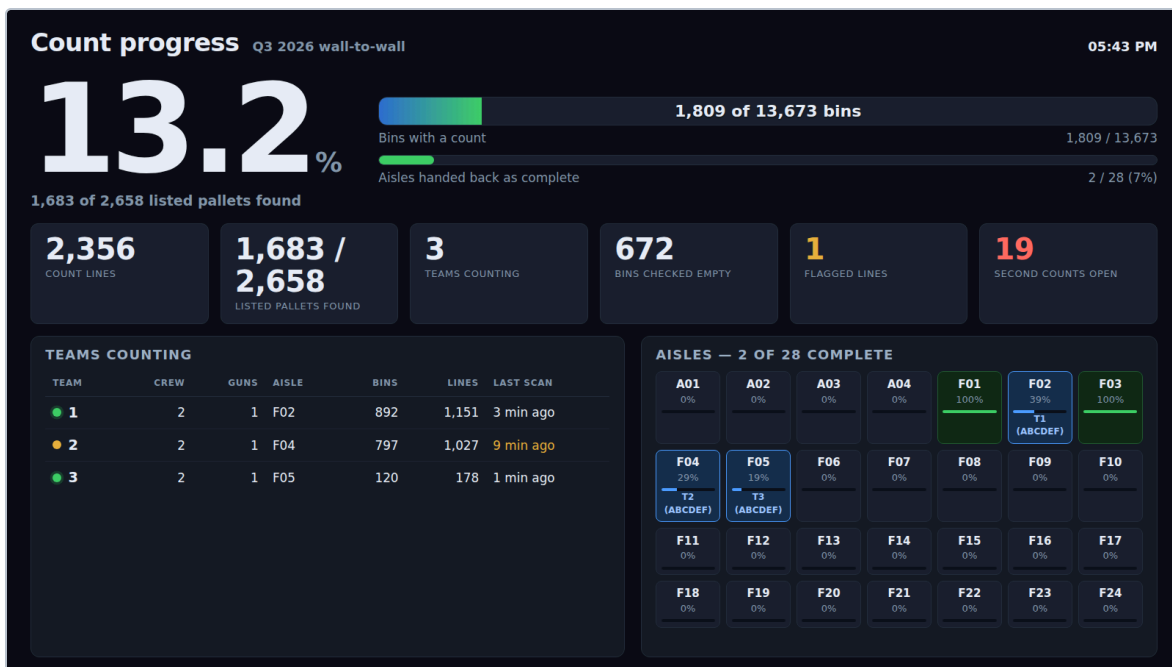
Watch for, during the count:

- Exceptions climbing on one team — usually a training problem, worth a radio call.
- A team stopped for a long time — dead battery, or stuck behind a trailer.
- An aisle nobody has started by mid-afternoon.

Step 17 — The office board

Put `/board` on the office TV. It needs no sign-in and is read-only: the percentage counted, bins with a count, aisles handed back as complete, a row per team with what they are on, and every aisle as a tile coloured by state. It deliberately shows no pallet IDs and no clock-in numbers.

It follows the newest open count on its own. To pin it to a particular one, add the session to the address: `/board?session=12`.



The office board, for the TV: percentage counted, aisles complete, every team, and every aisle coloured by state.

Part 5 — Finishing a count

Step 18 — Work the second counts

Dashboard → Second counts.

A second count is a bin somebody has to walk back to. They are raised automatically when a line disagrees with the inventory report (subject to your thresholds), or by a supervisor.

- **Raise from all variances now** goes through everything that currently disagrees and raises the tasks. It deliberately leaves alone pallets in aisles nobody has counted yet — mid-count those are not missing, just not reached — and tells you how many it skipped and why.
- The tasks appear on the handhelds: a team taps **Start second counts** on the assignment screen, counts the bin again, and presses **Bin done — nothing more here**.
- A second count that agrees with the first settles the line.

Work these down before you close the count. A count closed with open second counts is a count with known-wrong numbers in it.

Second counts 19 open · 0 done

A second count is a bin somebody goes back to. They are raised automatically when a first count disagrees with the inventory report (quantity differs, wrong bin, not on the report, counted twice), when an aisle is handed back with expected pallets nobody saw, or by hand here. The counter is told the bin and the reason type, never the numbers, and the team that did the first count is never offered the second. Lines from a second count replace the first for that bin.

BIN

F03B012

or

PALLET ID

PLT03012B

NOTE

why

TEAM (OPTIONAL)

any

Request second count

Raise from all variances now

Export CSV

BIN	PALLET	REASON	DETAIL	SOURCE	1ST TEAM	TEAM	STATUS	1ST COUNT	2ND COUNT
F01A005	F01-008	QTY VARIANCE	expected 40, first count 26	auto	1	any	open	F01-008 ×26.0	done ×
F01A021	F01-024	QTY VARIANCE	expected 32, first count 41	auto	1	any	open	F01-022 ×58.0, F01-023 ×25.0, F01-024 ×41.0	done ×
F01C107	F01-053	WRONG BIN	expected in F01A041, found in F01C107	auto	1	any	open	F01-053 ×35.0, EMPTY ×0.0	done ×
F02A005	F02-008	QTY VARIANCE	expected 40, first count 26	auto	1	any	open	F02-008 ×26.0	done ×
F02A021	F02-024	QTY VARIANCE	expected 32, first count 41	auto	1	any	open	F02-022 ×58.0, F02-023 ×25.0, F02-024 ×41.0	done ×
F02D031	F02-052	WRONG BIN	expected in F02A040, found in F02D031	auto	1	any	open	F02-052 ×28.0	done ×

Dashboard → Second counts. The counter is told the bin and the reason type, never the numbers — and never the team that counted it first.

Step 19 — Read the reports

Dashboard → Reports.

- **Pallet report** — one row per pallet: expected vs counted, which bin the ERP expected and where it was actually found, and a status you can act on: **MATCH**, **QTY VARIANCE**, **WRONG BIN**, **MISSING**, **NOT IN MASTER**, **COUNTED TWICE**. Lot and expiry get their own columns, because the right count of the wrong lot is still wrong. **Only exceptions** narrows it to what needs attention — including a wrong lot or a date about to run out, even when the quantity is right.

Pallet report

☒ Hide matching pallets **Pallet report CSV** **Raw counts CSV** **Exceptions CSV** **Uncounted bins CSV**

PALLET	SKU	DESCRIPTION	EXPECTED	1ST COUNT	COUNTED	EXPECTED BIN	FOUND IN	LOT	EXPIRY	TEAM	COMMENTS	STATUS
F01-008	SKU-6610	Salmon fillet skin-on 10lb	40		26	F01A005	F01A005	L2026152	2027-01-22	1		QTY VARIANCE
F01-011	SKU-3350	Hash brown patty 240ct	61		61	F01A008	F01A008	L2026191	2026-09-28 soon	1		MATCH
F01-017	SKU-4120	Chicken breast IQF 40lb	63		63	F01A011	F01A011	L2026230	2026-08-27 expired	1		MATCH
F01-022	SKU-6610	Salmon fillet skin-on 10lb	58		58	F01A021	F01A021	L2026282	2026-09-28 soon	1		MATCH
F01-024	SKU-3310	Fries shoestring 6x5lb	32		41	F01A021	F01A021	L2026282	2027-02-07	1		QTY VARIANCE

Showing 500 of 1233 — export the CSV for the full list.

The pallet report with **Hide matching pallets** ticked: a short pallet, an expired date, one expiring soon. This is the list to work down.

- **Find a lot** — after a recall notice: type part of a lot code and get every case of it, both where it was actually counted and where the report expected it, so a pallet nobody found still shows up.

Find a lot after a recall notice: where is every case of it

Searches what was actually counted, and what the inventory report expected — so it answers from the floor, not from what the ERP believed before the count started. Needs the lot code question turned on, or a report that carries lot codes.

LOT OR BATCH CODE

L2026139 **Find it**

15 counted, 18 on the report
Counted rows are where it actually is now.

PALLET	LOT	EXPIRY	QTY	BIN	AISLE	ITEM	TEAM	STATE
F01-005	L2026139	2027-01-19	59	F01A004	F01	SKU-2240 — Sweetcorn supersweet 20lb	1	done counted
F01-006	L2026139	2027-01-20	26	F01A004	F01	SKU-6610 — Salmon fillet skin-on 10lb	1	done counted
F01-007	L2026139	2027-01-21	33	F01A004	F01	SKU-6640 — Cod loin 8lb	1	done counted
F02-005	L2026139	2027-01-19	59	F02A004	F02	SKU-2240 — Sweetcorn supersweet 20lb	1	done counted
F02-006	L2026139	2027-01-20	26	F02A004	F02	SKU-6610 — Salmon fillet skin-on 10lb	1	done counted
F02-007	L2026139	2027-01-21	33	F02A004	F02	SKU-6640 — Cod loin 8lb	1	done counted

A recall: five cases of the lot found on the floor, and a sixth the report still expects in an aisle nobody has reached.

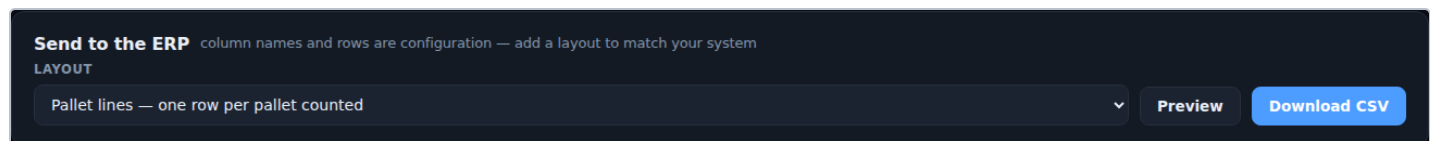
- **Count sheets** — printable paper sheets by aisle and level, for a dead battery or an auditor.

- **Exports** — the full count (every line as scanned), exceptions only, uncounted bins, the pallet report, second counts.

Step 20 — Send it to the ERP

Settings → ERP & backups → Send to the ERP.

1. Choose the **layout** that matches your ERP. Column names and which rows are included are configuration — add a layout rather than editing the file by hand afterwards.
2. **Preview** and read the first rows.
3. **Download CSV** and import it into the ERP.



Settings → ERP & backups. Preview before you download; the layout decides the column names and which rows are included.

Step 21 — Close the count

Dashboard → Progress → Count session → Close session. A closed count is read-only: scanners can no longer post lines to it, and its reports stay available for ever.

Delete... is separate and deliberately harder: the count must be closed first, you have to type its name, and a backup is taken before anything is removed. Everything counted against it goes with it. Close counts; delete only the ones created by mistake.

Step 22 — Backups and the log

Settings → ERP & backups → Backups & log.

- **Back up now** before and after anything large — an ERP import, a bulk re-upload, a deletion. The app also backs itself up daily and keeps the last 14.
- **The log** records every supervisor action: who signed in, who uploaded what, who overrode what, who raised second counts, who searched for a lot. **Export the log** for an auditor.

Backups & log

Back up now

Export the log

1 kept (newest 9/16/2026, 5:42:50 PM), one a day, 14 retained

BACKUP	SIZE	TAKEN		WHEN	WHO	DID WHAT
inventory-2026-09-16-17-42-50-daily.db	188 KB	9/16/2026, 5:42:50 PM	Download	9/16/2026, 5:42:54 PM	shared password	signed in with the shared password
				9/16/2026, 5:42:52 PM	Dana Whitfield	generated a cycle batch
				9/16/2026, 5:42:52 PM	Dana Whitfield	uploaded pallets
				9/16/2026, 5:42:52 PM	Dana Whitfield	uploaded bins
				9/16/2026, 5:42:52 PM	Dana Whitfield	created session
				9/16/2026, 5:42:52 PM	Dana Whitfield	created session
				9/16/2026, 5:42:52 PM	Dana Whitfield	raised second counts from the variances

Backups on the left, the log on the right. Both export.

Part 6 — The cycle-count programme

A cycle count is the same app with a different rhythm: instead of stopping the warehouse, a handful of bins are counted every day.

1. **Create one count session of type "Cycle count"** and keep using it. It runs for the year — do not create a new one each week, or you lose the history of when each bin was last counted.
2. Upload the bin list and the inventory report. If your ERP export has a *last physical inventory date* column it is read automatically, and the programme picks up where the ERP left off.
3. Go to `/cycle` → **Today's bins**. Choose:
 - **how many bins** in the batch,
 - **which bins**: *longest since it was counted, never counted, or a random sample*,
 - optionally a **zone, aisle or levels** to stay inside.
4. **Preview**, then **Generate**. The bins appear on the handhelds as tasks.

`/cycle` → *Today's bins: how many, which ones, and optionally the zone or aisles to stay inside.*

1. Counters pick **Cycle count** at sign-on and work the list. Each bin is counted as it stands — it is *the* count for that bin, not a second opinion.
2. **Coverage** shows how much of the warehouse has been counted in the period, and what has not been touched. Export it for the auditors.
3. A **schedule** generates the batch automatically each day or week so nobody has to remember.

Inventory Count

Front Royal

Dashboard

Cycle counts

Teams & crew

Settings

shared password

Log out

Cycle counts

America/New_York · 2026-09-16

Today

Still open

Coverage

Program & data

Program

a cycle-count session runs for the year — keep using the same one

CYCLE COUNT

#3 — Cycle count programme 2026

Close

NEW PROGRAM

e.g. 2026 cycle counts

Create

Bin list & inventory report

The bin list is uploaded once. Re-upload the **inventory report** whenever you want current expected quantities — tick *Replace existing* so old pallets are cleared out.

FILE

CSV OR EXCEL

Inventory report — pallet id, sku, description, qty,

Choose File

No file chosen

☒ Replace existing

Upload

Load Front Royal bin list

The batches that have been generated, and the programme's own bin list and inventory report.

Part 7 — When something goes wrong

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
Gun: <i>"this scanner is no longer authorised"</i>	Its link was reset or the scanner was removed	Settings → Scanners → Reset link , open the new link on the device
Gun: <i>"Offline and no list cached for this session"</i>	The handheld has never downloaded this count	Carry it into Wi-Fi once and sign on again
Gun says OFFLINE with lines queued	Normal in a dead spot	Nothing. They upload when it gets a signal. Do not wipe the device
A scan does nothing	DataWedge is not sending the Enter key	DataWedge → Basic data formatting → Send ENTER key
A scan seems to land on a button instead of the box	Something else took the focus	Nothing — the app puts the keystrokes in the box and carries on. Tell us if it still happens
The guide is a bin or two ahead of the team	The bins hold several pallets each	Fixed: the guide now stays on a bin until its tags are counted. Check Show the next bin in the aisle is on
A bar with the web address appears on every scan	The app is being run as a page in Chrome rather than installed, so Chrome shows its address bar whenever the page moves	Install it: Chrome menu → Add to Home screen , or the Install on this scanner button on the sign-on screen, then open it from the home-screen icon
The browser's address bar is in the way	Same thing — the app is not installed	As above. For a device that should run nothing else, use Zebra's Enterprise Home Screen
Gun: <i>"Team N is counting aisle X"</i>	Another team holds that racking block	Wait, or hand the other aisle back first
Second-count list is enormous	Thresholds are at 0	Set Recount over and or over % , and a cap (Part 3, step 13)
The map is a schematic, not your drawing	No rack drawing on this count	Dashboard → Progress → Count session → Map drawing

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
A reason code edit "did not reach" a gun	It takes up to about half a minute, and only lands between pallets	Wait for the counter to finish the pallet they are on
Can nobody sign in?	All logins lost	The shared password turns itself back on when there are no admin accounts; the superadmin login is recreated at start-up
Numbers look wrong after an ERP import	The inventory report moved on	Re-upload it with Replace what is there , then re-read the pallet report

Part 8 — Capacity: what the system will take

Measured against the real Front Royal bin list (13,673 bins), with fifteen teams and thirty scanners counting simultaneously while the office board and two dashboards refreshed over the top of them:

Scanners posting at once	30, no errors, nothing lost, nothing double-counted
Throughput	~1,200-1,600 count lines per second
A scan, under that full load	75 ms median, 205 ms at the 95th percentile
Office board / dashboard refresh	~140 ms median while all thirty guns were writing
Whole master list to a handheld	~0.4 s each, thirty pulling simultaneously

Thirty scanners is comfortably inside what the server will take — real counters scan a pallet every ten to twenty seconds, which is a small fraction of the load above. The constraints that matter in practice are warehouse Wi-Fi coverage and battery life, not the server.

This is checked automatically: `node tests/run-all.mjs load-15-teams` runs the whole scenario and fails if anything is lost, doubled, mis-attributed, or if two teams ever end up in one racking block.

Appendix A — File column reference

Files may be CSV, TSV or Excel (.xlsx/.xlsm). Column names are matched loosely — case, spaces and punctuation are ignored — so most ERP exports load unchanged. Each upload card shows exactly which columns it recognised.

Bin list — the only required column is the location.

WHAT	ACCEPTED COLUMN NAMES
Bin (required)	location, loc, bin, bin location, location code, slot, code, warehouse location
Aisle	aisle, row, aisle no, aisle number — <i>derived from the bin code if absent</i>
Zone	zone, area, section, region, warehouse
Level	level, levels, tier, shelf — <i>derived from the bin code if absent</i>
Description	description, desc, name
Last counted	last phys invt date, last counted, last count date, last inventory date

Inventory report — the only required column is the pallet ID.

WHAT	ACCEPTED COLUMN NAMES
Pallet (required)	pallet id, pallet, container, container id, LPN, licence plate, pallet no, id, tag
SKU	sku, item, item number, item code, part number, product, material, stock code
Description	description, desc, item description, product name
Quantity	qty, quantity, on hand, on hand qty, expected, expected qty, system qty, cases, units
Unit	uom, unit, unit of measure, um
Location	<i>same names as the bin list</i>
Lot	lot, lot code, lot no, lot number, batch, batch code, batch number
Expiry	expiry, expiry date, expiration, expires, best before, use by, shelf life date

Counting plan — one row per aisle, in the order the team counts it.

WHAT	ACCEPTED COLUMN NAMES
Team	team, team number, crew, group
Aisle	aisle, row, aisle no
Levels	level, levels, tier, shelf — e.g. <code>A-C</code> , <code>D-F</code> , <code>A-F</code>

Dates are read in either order — `2027-03-15` and `03/15/2027` both work.

Appendix B — Server settings reference

SETTING	DEFAULT	WHAT IT DOES
<code>ADMIN_PASSWORD</code>	<code>changeme</code>	The shared supervisor password. Change it
<code>SHARED_PASSWORD_LOGIN</code>	<code>on</code>	<code>off</code> requires named logins — ignored if no admin account exists, so it cannot lock you out
<code>SUPERADMIN_USER</code>	—	Username of the permanent admin, created at start-up
<code>SUPERADMIN_NAME</code>	—	That person's name, as it appears in the log
<code>SUPERADMIN_PASSWORD</code>	falls back to <code>ADMIN_PASSWORD</code>	Must be 8+ characters and not <code>changeme</code>
<code>SCANNER_AUTH</code>	<code>on</code>	<code>off</code> lets anything on the network post counts. Leave it on
<code>DB_PATH</code>	<code>data/inventory.db</code>	Put this on a persistent volume
<code>BACKUP_DIR</code>	<code>data/backups</code>	Where the daily backups go
<code>BACKUP_KEEP</code>	14	How many backups to keep
<code>SITE_TIMEZONE</code>	<code>America/New_York</code>	Dates on reports and cycle-count due dates
<code>PORT</code> / <code>HOST</code>	3000 / 0.0.0.0	Where the server listens
<code>MAX_UPLOAD_MB</code>	64	Largest file an upload will accept